| Question | | | Marking details | Marks available | | | | Maths | Prac |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | | |
| | (c) | (i) | Centripetal force = 0.010 $\frac{v^2}{R}$<br>(1 for $\frac{mv^2}{R}$; 1 if value inserted for $m$) | 1 | 1 | | 2 | 1 | 2 |
| | | (ii) | Forces acting on mass Q: $0.090\ g - \tau = 0$ $\quad \tau$: tension (1)<br>So $\tau = 0.090g$. Substitution for $\tau$ into (c)(i) (1)<br>$0.090g = 0.010\dfrac{v^2}{R}$<br>$v^2 = \dfrac{0.090g}{0.010}\ R$<br>$v^2 = 9g\ R$ clear and convincing working (1) | | 3 | | 3 | 2 | 3 |